**Reverse- and Reengineering**
**Homework 4**

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This sheet is not for submission. However, we strongly encourage you to solve the problems and bring your solutions to the tutorials for discussion.

## 1 x86-64 Assembly

Consider the x86-64 assembly function `decode` below. Write an equivalent C function with the signature `void decode(long *xp, long *yp)` that performs the exact same operations.

```
decode:
        push   rbp
        mov    rbp, rsp
        mov    qword ptr [rbp - 8], rdi
        mov    qword ptr [rbp - 16], rsi
        mov    rax, qword ptr [rbp - 8]
        mov    rax, qword ptr [rax]
        mov    qword ptr [rbp - 24], rax
        mov    rax, qword ptr [rbp - 16]
        mov    rax, qword ptr [rax]
        mov    qword ptr [rbp - 32], rax
        mov    rcx, qword ptr [rbp - 24]
        mov    rax, qword ptr [rbp - 16]
        mov    qword ptr [rax], rcx
        mov    rcx, qword ptr [rbp - 32]
        mov    rax, qword ptr [rbp - 8]
        mov    qword ptr [rax], rcx
        pop    rbp
        ret
```

## 2 Control-Flow Flattening

Consider the following C function `bar`:

```c
int bar(){
    int i=1;
    int s=0;
    while (i <= 100) {
        s+=i;
        i++;
    }
    return s;
}
```

1. Apply the control-flow flattening technique to transform this code.
2. Draw the corresponding flattened control-flow graph (CFG).